Sign Test and Wilcoxon Signed-Rank Test

Presentation Outline

1. Inference for Paired Data Review and Assumptions
2. Sign Test
3. Wilcoxon Signed-Rank Test

Review: t-test for Paired Data

Compare observations of two similar quantitative variables from same cases or from matched cases.

Method: Hypothesis Test and Confidence Interval for population mean difference μ_D.

Based on Sample Mean Difference $\overline{D}$

- Sample Mean Difference is highly affected by outliers
- You must have values for all observations to calculate sample mean difference.

This method may not be appropriate if you have outliers or other data issues.

Assumptions: t-test for Paired Data

Sampling distribution of sample mean difference $\overline{D}$ is a Normal distribution

- Sample size may not be large enough for this assumption to be true.

This method may not be appropriate if above assumption is not met by your data.

Sign Test

Compare observations of two similar quantitative variables from same cases or from matched cases.

Method is based on population median difference instead of population mean difference.

H_0: Population Median difference equals 0.

H_a: Population Median difference is not equal to 0.

Sign Test Statistic and p-value

Calculate $D_i =$ difference between two values for i^{th} case.

Test Statistic: $S =$ number of positive observations of D_i

If $S \geq \frac{n}{2}$, p-value is:

$$2 * 0.5^n * \sum_{x=S}^{n} \binom{n}{x}$$

If $S < \frac{n}{2}$, p-value is:

$$2 * 0.5^n * \sum_{x=0}^{S} \binom{n}{x}$$

Example - Sign Test

Twin Schizophrenia Data

- Unaffect = left hippocampus size for twin without schizophrenia
- Affected = left hippocampus size for twin with schizophrenia

H_0: Population Median difference in left hippocampus size between twin without schizophrenia and twin with Schizophrenia is 0.

H_a: Population Median difference in left hippocampus size between twin without Schizophrenia and twin with Schizophrenia is not equal to 0.

Example - Sign Test

Out of 15 observations, $S = 14$ have a positive value of D_i = Unaffect - Affected

Since $S \geq \frac{n}{2} = \frac{15}{2}$, p-value is:

$$2 * 0.5^{15} * \sum_{x=14}^{15} \binom{15}{x} = 0.00098$$

Conclusion:

We have extremely strong evidence the population median difference in left hippocampus size between twin without Schizophrenia and twin with Schizophrenia is not equal to 0.

Wilcoxon Signed-Rank Test

Compare observations of two similar quantitative variables from same cases or from matched cases.

Method is based on population median difference instead of population mean difference.

Used when differences D_i are symmetric

More Power than Sign Test for detecting positive or negative population median differences.

H_0: Population Median difference equals 0.

H_a: Population Median difference does not equal 0.

Wilcoxon Signed-Rank Test Statistic and *p*-value

Steps for calculating Test Statistic:

1. Calculate D_i = difference between two values for i^{th} case.
2. Rank $|D_i|$ = absolute difference between two values. Assign the average of the ranks for ties.
3. Assign $\text{sign}(D_i)$ to these ranks, so positive differences have positive ranks and negative differences have negative ranks.
4. Test Statistic: V = sum of ranks for positive differences

p-value: Obtained by comparing V to special distribution of the Wilcoxon Signed Rank Test Statistic.

Example - Wilcoxon Signed-Rank Test

Twin Schizophrenia Data

- Unaffect = left hippocampus size for twin without schizophrenia
- Affected = left hippocampus size for twin with schizophrenia

H_0: Population Median difference in left hippocampus size between twin without schizophrenia and twin with Schizophrenia is 0.

H_a: Population Median difference in left hippocampus size between twin without Schizophrenia and twin with Schizophrenia is not equal to 0.

Example - Wilcoxon Signed Rank Test

14 out of 15 differences are positive.

In absolute value, the only negative difference has a rank of 9.

Sum of ranks with positive differences: $V = 111$

p-value $= 0.0020$

Conclusion: We have very strong evidence the population median difference in left hippocampus size between twin without Schizophrenia and twin with Schizophrenia is not equal to 0.